

SOF IMO — Topic Deep-Dive: Time & Money

For Ambar, Class III CBSE

PART A — TIME

1. Reading the clock

- The **short hand** shows the HOUR; the **long hand** shows the MINUTES.
- Each number the minute hand passes = 5 minutes.

Minute hand at	We say	Digital
12	o'clock	3:00
3	quarter past	3:15
6	half past	3:30
9	quarter to (the NEXT hour)	3:45 = quarter to 4

The “quarter to” trap: 5:45 is quarter to **6**, not quarter to 5. After the half-hour, we name the hour that is COMING.

2. Units of time

Fact	Value
1 hour	60 minutes
Half an hour	30 minutes
Quarter of an hour	15 minutes
1 day	24 hours
1 week	7 days
1 year	12 months, 365 days (366 in a leap year)

Month lengths: knuckle trick — Jan 31, Feb 28/29, Mar 31, Apr 30, May 31, Jun 30, Jul 31, Aug 31, Sep 30, Oct 31, Nov 30, Dec 31. **February is the shortest.**

3. Working with time

Duration: count on from start to end. 7:30 → 9:30 is 2 hours. 4:15 → 5:00 is 45 minutes (count 4:15 → 4:30 → 4:45 → 5:00). **End time:** start + duration. Match starts at 10:00, lasts 3 and a half hours → 10:00 + 3 h = 1:00, + 30 min = **1:30 (half past 1)**. **Calendar days:** “If 15 August is Friday, 17 August is?” — count 16 Sat, 17 **Sunday**. “After 3 days from Monday” → Tue, Wed, **Thursday**. **Weeks to days:** 2 weeks 3 days = 14 + 3 = **17 days**.

PART B — MONEY

4. Rupees and paise

- ₹1 = 100 paise.** So ₹2 = 200 paise; 50 paise + 50 paise = ₹1.
- Coins: ₹1, ₹2, ₹5, ₹10. Notes: ₹10, ₹20, ₹50, ₹100, ₹200, ₹500.
- Writing: ₹5.50 means 5 rupees and 50 paise.

5. Adding money and making change

Adding: line up the rupee point. **Worked example:** ₹5.50 + ₹3.75 → 50 p + 75 p = 125 p = ₹1.25 → ₹5 + ₹3 + ₹1.25 = **₹9.25**.

Change = money given – cost. **Worked example:** 4 samosas at ₹8 each = ₹32. Paid with ₹50 → change = 50 – 32 = **₹18**. (Two steps: multiply, then subtract.)

Counting mixed money: 3 coins of ₹5 + 2 notes of ₹10 = 15 + 20 = **₹35**.

6. Common traps

Trap	Wrong	Right
Quarter to 6 read as 5 :15	5:15	5:45
1 hour = 100 minutes	100	60 minutes
Counting the start day	Monday + 3 days = Wednesday	Thursday
Answering total spent when asked change	paid 50 for 32 → "32"	change ₹18
₹5.50 + ₹3.75 without carrying paise	₹8.25	₹9.25 (125 p = ₹1.25)
Hour hand vs minute hand mixed up	long hand = hours	short = hour, long = minutes

7. High-frequency IMO question types

1. Read the clock: o'clock, half past, quarter past/to.
2. Conversions: hours-minutes, weeks-days, rupee-paise.
3. Duration and end time ("movie starts at 4, lasts 2 hours").
4. Calendar: day after N days, days in a month, months order.
5. Total cost of several items; total of coins and notes.
6. Change from ₹10/₹20/₹50/₹100.
7. Achiever's: two purchases then change; start time + hours = clock phrase.

8. Key facts checklist

- ☐ 1 h = 60 min; half hour = 30 min; quarter hour = 15 min
- ☐ Quarter past = :15; half past = :30; quarter to = :45 of the NEXT hour
- ☐ 1 day = 24 h; 1 week = 7 days; 1 year = 12 months = 365 days
- ☐ February is the shortest month
- ☐ ₹1 = 100 paise
- ☐ Change = amount given – total cost

9. Quick drill (10 questions)

1. 3 hours = ____ minutes
2. Quarter past 8 = ____ (digital)
3. 6:45 in words = ?
4. If today is Wednesday, what day is it after 2 days?
5. 3 weeks = ____ days
6. ₹3 = ____ paise
7. ₹45 + ₹25 = ?
8. A toy costs ₹36. Ambar pays ₹50. Change = ?
9. School starts at 8:00 and the first break is at 10:30. How long until the break?
10. 2 notes of ₹20 and 3 coins of ₹5 = ?

Answers: 1) 180 2) 8:15 3) quarter to 7 4) Friday 5) 21 6) 300 7) ₹70 8) ₹14 9) 2 and a half hours 10) ₹55 (40 + 15)